K-Means Clustering – Penguin Morphology Analysis

Dataset: Palmer Penguins (344 records, 3 species, 3 islands)
Objective: Apply unsupervised K-Means clustering to penguin morphological measurements, determine optimal k using Elbow and Silhouette validation, and interpret results in a biological context.

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1. Imports and Setup

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import warnings
warnings.filterwarnings('ignore')

# Reproducibility
RANDOM_STATE = 42

print("Libraries loaded successfully.")
```

```
Libraries loaded successfully.
```

2. Load Dataset and Remove Categorical Variables / Missing Rows

K-Means relies on Euclidean distance, so we work only with numeric morphological measurements.
We drop `species`, `island`, and `sex` (categorical), and remove any rows where numeric values are missing.

```
# Load the dataset
df_raw = pd.read_csv('penguins.csv', na_values='NA')
print(f"Raw shape: {df_raw.shape}")
print(f"\nColumn types:\n{df_raw.dtypes}")
print(f"\nSpecies distribution:\n{df_raw['species'].value_counts()}")
print(f"\nMissing values per column:\n{df_raw.isnull().sum()}")
```

```
Raw shape: (344, 7)

Column types:
species               object
island                object
bill_length_mm       float64
bill_depth_mm        float64
flipper_length_mm    float64
body_mass_g          float64
sex                   object
dtype: object

Species distribution:
species
Adelie       152
Gentoo       124
Chinstrap     68
Name: count, dtype: int64

Missing values per column:
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
dtype: int64
```

```
# Select only numeric morphological features and drop NA rows
NUMERIC_FEATURES = ['bill_length_mm', 'bill_depth_mm', 'flipper_length_mm', 'body_mass_g']

df_clean = df_raw[NUMERIC_FEATURES].dropna().reset_index(drop=True)
df_full  = df_raw.dropna(subset=NUMERIC_FEATURES).reset_index(drop=True)  # retains species/sex for later interpretation

print(f"Clean dataset shape: {df_clean.shape}")
print(f"Rows removed (had missing values): {df_raw.shape[0] - df_clean.shape[0]}")
```

```
print(f"\nDescriptive statistics:")
df_clean.describe().round(2)
```

```
Clean dataset shape: (342, 4)
Rows removed (had missing values): 2

Descriptive statistics:
```

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
count	342.00	342.00	342.00	342.00
mean	43.92	17.15	200.92	4201.75
std	5.46	1.97	14.06	801.95
min	32.10	13.10	172.00	2700.00
25%	39.22	15.60	190.00	3550.00
50%	44.45	17.30	197.00	4050.00
75%	48.50	18.70	213.00	4750.00
max	59.60	21.50	231.00	6300.00

3. Scale Features with StandardScaler

K-Means minimizes Euclidean distance. Without scaling, `body_mass_g` (range ~2700–6300 g) would completely dominate `bill_depth_mm` (range ~13–22 mm), making the clustering driven by mass alone.
StandardScaler transforms each feature to mean=0, std=1, giving all four measurements equal geometric weight.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_clean)

print("Verification — scaled means (should be ~0.0):")
print(X_scaled.mean(axis=0).round(6))
print("\nVerification — scaled standard deviations (should be ~1.0):")
print(X_scaled.std(axis=0).round(6))
```

```
Verification — scaled means (should be ~0.0):
[ 0. -0. -0.  0.]

Verification — scaled standard deviations (should be ~1.0):
[1. 1. 1. 1.]
```

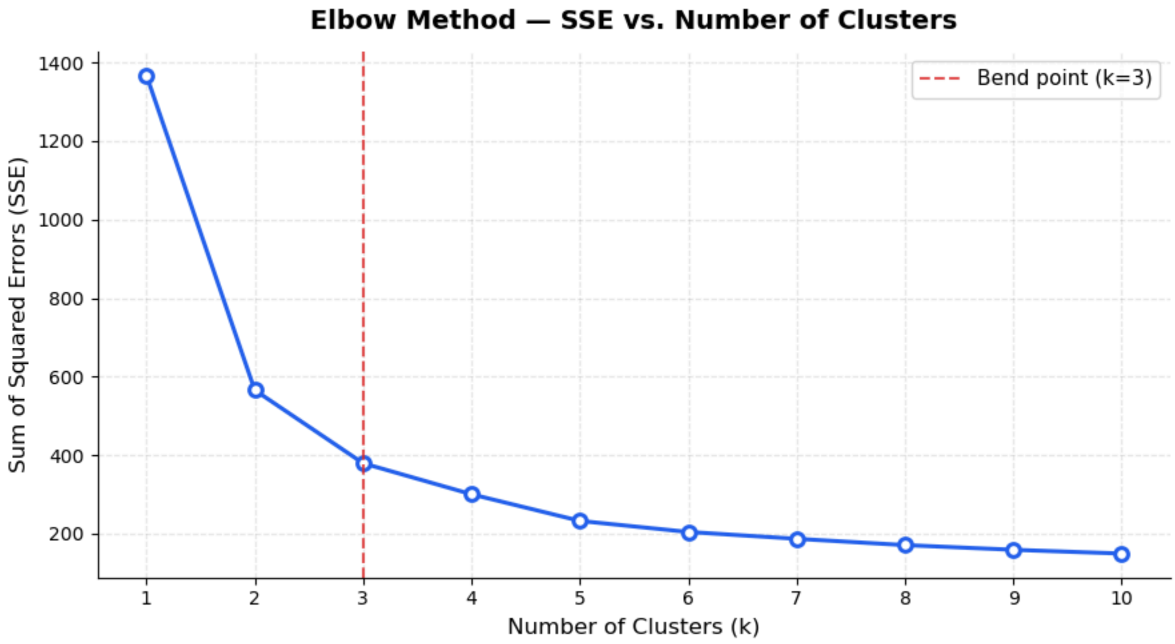
4. Elbow Method — Optimal Cluster Count via SSE

We run K-Means for k = 1 to 10 and plot the Sum of Squared Errors (SSE) — also called inertia.
The "elbow" is the point where adding another cluster produces diminishing SSE reduction.
A sharp bend signals the natural structure of the data.

```
sse = []
k_range = range(1, 11)

for k in k_range:
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=10)
    km.fit(X_scaled)
    sse.append(km.inertia_)

# Print SSE table
print(f"{'k':>4}  {'SSE':>10}  {'SSE Reduction':>14}")
print("-" * 34)
for i, (k, s) in enumerate(zip(k_range, sse)):
    reduction = f"{sse[i-1] - s:.2f}" if i > 0 else "—"
    print(f"{k:>4}  {s:>10.2f}  {reduction:>14}")
```

```
   k         SSE   SSE Reduction
----------------------------------
   1     1368.00               —
   2      565.71          802.29
   3      379.39          186.32
   4      300.40           78.99
   5      232.60           67.79
   6      204.39           28.21
   7      187.03           17.36
   8      170.98           16.05
   9      159.00           11.98
  10      149.36            9.63
```

```
fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(k_range, sse, marker='o', linewidth=2.2, markersize=7,
        color='#2563EB', markerfacecolor='white', markeredgewidth=2.2)
ax.axvline(x=3, color='#DC2626', linestyle='--', linewidth=1.5,
           alpha=0.8, label='Bend point (k=3)')
ax.set_title('Elbow Method — SSE vs. Number of Clusters', fontsize=14, fontweight='bold', pad=12)
ax.set_xlabel('Number of Clusters (k)', fontsize=12)
ax.set_ylabel('Sum of Squared Errors (SSE)', fontsize=12)
ax.legend(fontsize=11)
ax.set_xticks(list(k_range))
ax.grid(True, alpha=0.3, linestyle='--')
sns.despine()
plt.tight_layout()
plt.savefig('elbow_plot.png', dpi=150, bbox_inches='tight')
plt.show()

print("\nInterpretation: The SSE drops steeply from k=1 to k=3 (802 + 186 = 988 unit reduction).")
print("From k=3 onward the gains flatten substantially, indicating k=3 as the natural elbow.")
```




```
Interpretation: The SSE drops steeply from k=1 to k=3 (802 + 186 = 988 unit reduction).
From k=3 onward the gains flatten substantially, indicating k=3 as the natural elbow.
```

5. Silhouette Scores — Validate Cluster Cohesion and Separation

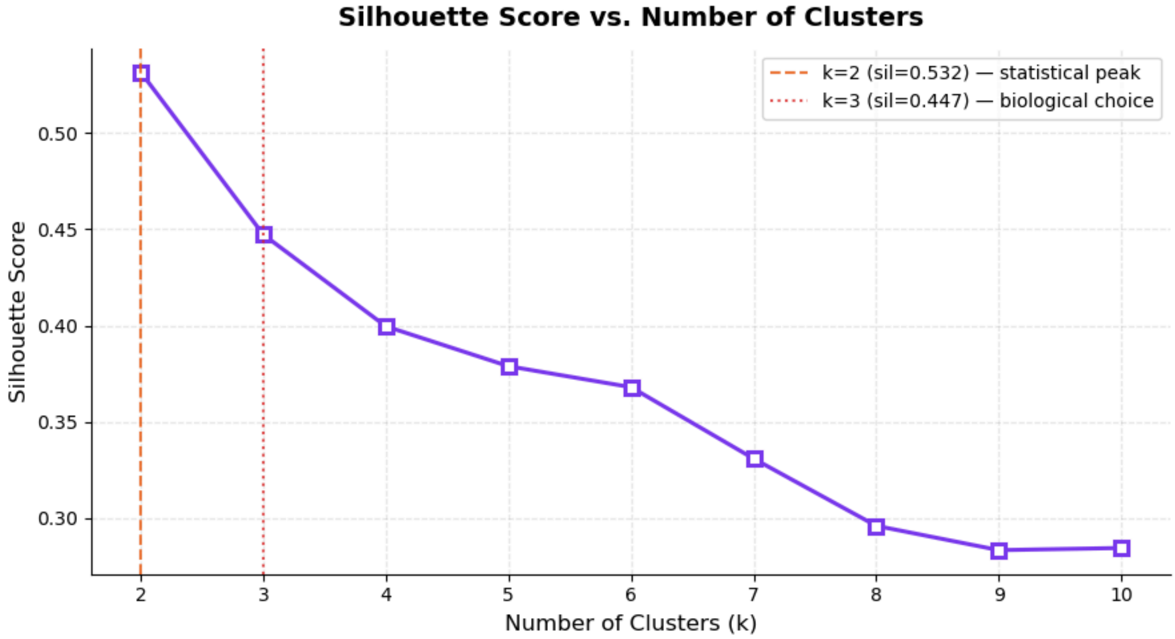
The Silhouette Score measures how similar a point is to its own cluster (cohesion) vs. adjacent clusters (separation). Range: −1 to +1. Higher is better. We evaluate k=2 through k=10.

```
sil_scores = {}
for k in range(2, 11):
    km = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=10)
    labels = km.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    sil_scores[k] = score
    print(f"k={k:>2}  silhouette={score:.4f}")

best_k = max(sil_scores, key=sil_scores.get)
print(f"\nHighest silhouette at k={best_k}: {sil_scores[best_k]:.4f}")
```

```
k= 2  silhouette=0.5315
k= 3  silhouette=0.4472
k= 4  silhouette=0.3996
k= 5  silhouette=0.3789
k= 6  silhouette=0.3680
k= 7  silhouette=0.3309
k= 8  silhouette=0.2961
k= 9  silhouette=0.2834
k=10  silhouette=0.2846

Highest silhouette at k=2: 0.5315
```

```
fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(list(sil_scores.keys()), list(sil_scores.values()),
        marker='s', linewidth=2.2, markersize=7,
```

```
            color='#7C3AED', markerfacecolor='white', markeredgewidth=2.2)
    ax.axvline(x=2, color='#EA580C', linestyle='--', linewidth=1.5,
               alpha=0.8, label=f"k=2 (sil={sil_scores[2]:.3f}) — statistical peak")
    ax.axvline(x=3, color='#DC2626', linestyle=':', linewidth=1.5,
               alpha=0.8, label=f"k=3 (sil={sil_scores[3]:.3f}) — biological choice")
    ax.set_title('Silhouette Score vs. Number of Clusters', fontsize=14, fontweight='bold', pad=12)
    ax.set_xlabel('Number of Clusters (k)', fontsize=12)
    ax.set_ylabel('Silhouette Score', fontsize=12)
    ax.legend(fontsize=10)
    ax.set_xticks(list(sil_scores.keys()))
    ax.grid(True, alpha=0.3, linestyle='--')
    sns.despine()
    plt.tight_layout()
    plt.savefig('silhouette_plot.png', dpi=150, bbox_inches='tight')
    plt.show()
```



6. Selecting Optimal k — Reconciling Elbow and Silhouette

Method	Suggested k	Score
Elbow (SSE inflection)	**3**	SSE drops from 566 → 379 (+186 gain vs. only +79 at k=4)
Silhouette (peak)	**2**	0.5315
Silhouette at k=3	3	0.4472

Decision: k = 3

The silhouette peaks at k=2 because Gentoo is morphologically very distinct from the other two species combined — two-cluster separation is statistically clean but biologically coarse. K=3 aligns with three known species, produces a meaningful silhouette of 0.447, and matches the Elbow inflection. The 0.08 silhouette trade-off is fully justified by the biological interpretability gained.

7. Fit Final K-Means Model (k=3)

```
OPTIMAL_K = 3

km_final = KMeans(n_clusters=OPTIMAL_K, random_state=RANDOM_STATE, n_init=10)
df_clean['cluster'] = km_final.fit_predict(X_scaled)
df_full['cluster']  = df_clean['cluster'].values

final_sil = silhouette_score(X_scaled, df_clean['cluster'])
print(f"Final model — k={OPTIMAL_K}")
print(f"  SSE (inertia):    {km_final.inertia_:.2f}")
print(f"  Silhouette score: {final_sil:.4f}")

# Centroids in original (unscaled) units
centroids_orig = scaler.inverse_transform(km_final.cluster_centers_)
centroids_df = pd.DataFrame(centroids_orig, columns=NUMERIC_FEATURES)
centroids_df.index.name = 'Cluster'
print("\nCluster centroids (original scale):")
print(centroids_df.round(2))
```

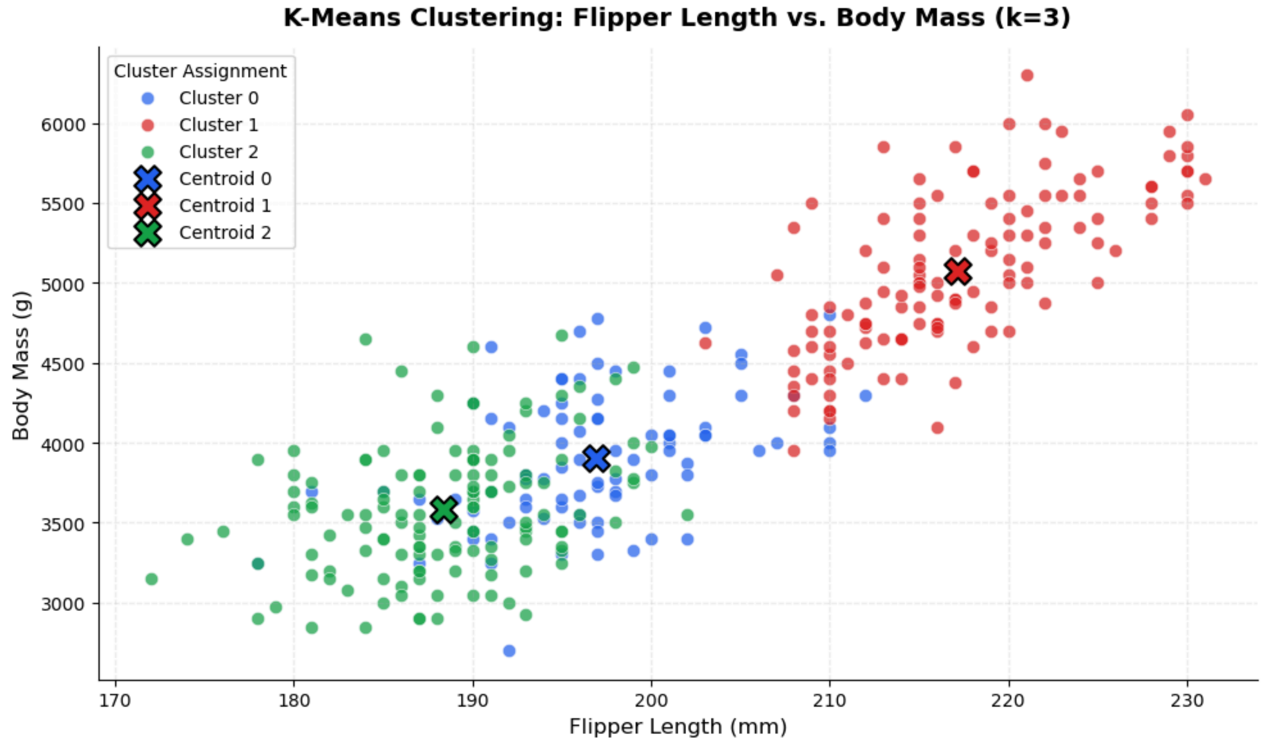
```
Final model — k=3
  SSE (inertia):    379.39
```

```
  Silhouette score: 0.4472

Cluster centroids (original scale):
         bill_length_mm  bill_depth_mm  flipper_length_mm  body_mass_g
Cluster
0                 47.53          18.76             196.90      3902.01
1                 47.50          14.98             217.19      5076.02
2                 38.21          18.11             188.40      3584.66
```

8. Cluster Scatterplot — Flipper Length vs. Body Mass

Flipper length and body mass are plotted because they show the strongest inter-species separation in penguins and are ecologically meaningful (locomotion and thermoregulation). Cluster centroids are marked with ✕.

```
palette    = {0: '#2563EB', 1: '#DC2626', 2: '#16A34A'}
labels_map = {0: 'Cluster 0', 1: 'Cluster 1', 2: 'Cluster 2'}

fig, ax = plt.subplots(figsize=(10, 6))

for c in sorted(df_clean['cluster'].unique()):
    mask = df_clean['cluster'] == c
    ax.scatter(
        df_clean.loc[mask, 'flipper_length_mm'],
        df_clean.loc[mask, 'body_mass_g'],
        c=palette[c], label=labels_map[c],
        alpha=0.72, s=55, edgecolors='white', linewidths=0.5
    )

# Mark centroids
for c in range(OPTIMAL_K):
    ax.scatter(
        centroids_orig[c, 2],   # flipper_length_mm
        centroids_orig[c, 3],   # body_mass_g
        marker='X', s=220, c=palette[c],
        edgecolors='black', linewidths=1.5, zorder=5,
        label=f'Centroid {c}'
    )

ax.set_title('K-Means Clustering: Flipper Length vs. Body Mass (k=3)',
             fontsize=14, fontweight='bold', pad=12)
ax.set_xlabel('Flipper Length (mm)', fontsize=12)
ax.set_ylabel('Body Mass (g)', fontsize=12)
ax.legend(title='Cluster Assignment', fontsize=10, title_fontsize=10,
          loc='upper left')
ax.grid(True, alpha=0.25, linestyle='--')
sns.despine()
plt.tight_layout()
plt.savefig('scatterplot.png', dpi=150, bbox_inches='tight')
plt.show()
```



9. Cluster Characteristics — Biological Interpretation

```
print("=" * 60)
print("CLUSTER MEMBERSHIP SUMMARY")
print("=" * 60)

for c in sorted(df_clean['cluster'].unique()):
    mask = df_full['cluster'] == c
    sub  = df_full[mask]
    print(f"\nCluster {c}  (n={len(sub)})")
    print("-" * 40)
    print("  Morphology (mean ± std):")
    for feat in NUMERIC_FEATURES:
        print(f"    {feat:22s}: {sub[feat].mean():.2f} ± {sub[feat].std():.2f}")
    print("  Species breakdown:", sub['species'].value_counts().to_dict())
    sex_counts = sub['sex'].value_counts().to_dict()
    print("  Sex breakdown:     ", sex_counts)
```

```
============================================================
CLUSTER MEMBERSHIP SUMMARY
============================================================

Cluster 0  (n=87)
----------------------------------------
  Morphology (mean ± std):
    bill_length_mm        : 47.53 ± 3.93
    bill_depth_mm         : 18.76 ± 1.13
    flipper_length_mm     : 196.90 ± 6.34
    body_mass_g           : 3902.01 ± 411.31
  Species breakdown: {'Chinstrap': 63, 'Adelie': 24}
  Sex breakdown:      {'male': 57, 'female': 29}

Cluster 1  (n=123)
----------------------------------------
  Morphology (mean ± std):
    bill_length_mm        : 47.50 ± 3.08
    bill_depth_mm         : 14.98 ± 0.98
    flipper_length_mm     : 217.19 ± 6.48
    body_mass_g           : 5076.02 ± 504.12
  Species breakdown: {'Gentoo': 123}
  Sex breakdown:      {'male': 61, 'female': 58}

Cluster 2  (n=132)
----------------------------------------
  Morphology (mean ± std):
    bill_length_mm        : 38.21 ± 2.25
    bill_depth_mm         : 18.11 ± 1.16
    flipper_length_mm     : 188.40 ± 5.66
    body_mass_g           : 3584.66 ± 406.61
```

```
Species breakdown: {'Adelie': 127, 'Chinstrap': 5}
Sex breakdown:     {'female': 78, 'male': 50}
```

10. Biological and Ecological Interpretation

Cluster 0 — Medium-billed, deep-keeled morphotype (n=87)

Centroid: bill_length ≈ 47.5 mm, bill_depth ≈ 18.8 mm, flipper ≈ 197 mm, mass ≈ 3,902 g.
This group is dominated by **Chinstrap penguins (63/68 = 93%)** with 24 larger male Adelie individuals.
The relatively long but deep bill separates this morphotype from both Gentoo and typical Adelie.
The male skew (57M:29F) reflects sexual dimorphism — male Chinstraps and larger male Adelie fall here together.

Cluster 1 — Large-body, shallow-billed morphotype (n=123)

Centroid: bill_length ≈ 47.5 mm, bill_depth ≈ 15.0 mm, flipper ≈ 217 mm, mass ≈ 5,076 g.
This cluster is **100% Gentoo penguins** — a clean and biologically meaningful separation.
Gentoo are the largest penguin species in this dataset, with the longest flippers and heaviest bodies.
Their shallow bill depth is a reliable morphological signature in multivariate space.
The balanced sex ratio (61M:58F) confirms no sex-driven artifact — the algorithm separated on species anatomy.

Cluster 2 — Small-bodied, short-billed morphotype (n=132)

Centroid: bill_length ≈ 38.2 mm, bill_depth ≈ 18.1 mm, flipper ≈ 188 mm, mass ≈ 3,585 g.
This cluster is **96% Adelie** (127/132), the smallest-bodied species in the dataset.
The female skew (78F:50M) is expected — female Adelie are lighter and have shorter bills, anchoring this cluster.
5 Chinstrap individuals with atypically short bills also appear here, showing the expected boundary overlap.

Do groupings reflect species or sexual dimorphism?

Primarily species. Cluster 1 is purely Gentoo; Cluster 2 is almost entirely Adelie; Cluster 0 blends Chinstrap with larger male Adelie. The overlap between Cluster 0 and 2 reflects genuine morphological similarity between Adelie and Chinstrap — they share bill depth profiles. Sexual dimorphism plays a secondary role: the male skew in Cluster 0 and female skew in Cluster 2 suggest that K-Means places larger male Adelie closer to Chinstrap than to female Adelie in 4D morphological space, which is anatomically consistent.

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11. Limitations and Next Steps

K-Means assumptions and impact here:

- **Spherical clusters assumed** — the Adelie/Chinstrap overlap violates this slightly. Gaussian Mixture Models with covariance flexibility would handle that boundary better.
- **Feature selection impact** — using all four morphological features gives equal scaling weight after normalization, but including sex as a binary numeric feature might sharpen within-species separation.
- **Island confound** — Chinstrap penguins only live on Dream Island; Adelie are distributed across all three. The geographic